



Getting Started with Blueshift

Introduction to Blueshift 2.0 Platform



blueshift

What is Blueshift

Systematic Trading Strategies – For Everyone!

IP protection



Strategy research



Financial datasets



Learning and community



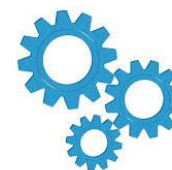
blueshift



On the cloud



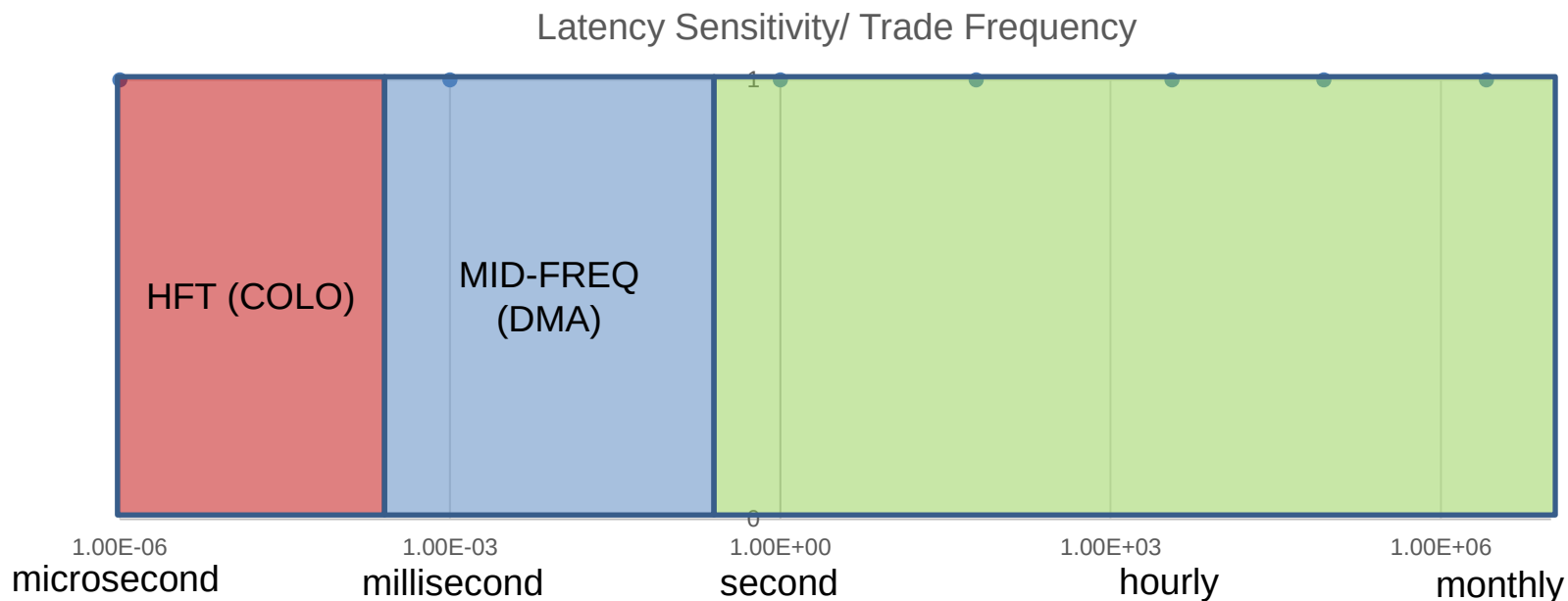
Live trading



Back-testing

A few definitions

What is high frequency trading? What is mid-frequency? What is low frequency?



Latency: Roundtrip time from an event to completed reaction

Blueshift is a low-frequency platform

Why Blueshift



1. Event-driven System

No **look-ahead** bias, consistent and **reusable** code between backtesting and live trading

2. Complex Event Processing (CEP) Engine

Not just a stateless rule-engine – can handle complex state-dependent logic

3. High Quality Data

Avoid garbage-in/garbage-out. Curated **survivorship bias free** datasets. **Fully adjusted** data. All historical data query are adjusted for **Point-in-time** (on-the-fly data adjustments)

4. Institutional Grade APIs

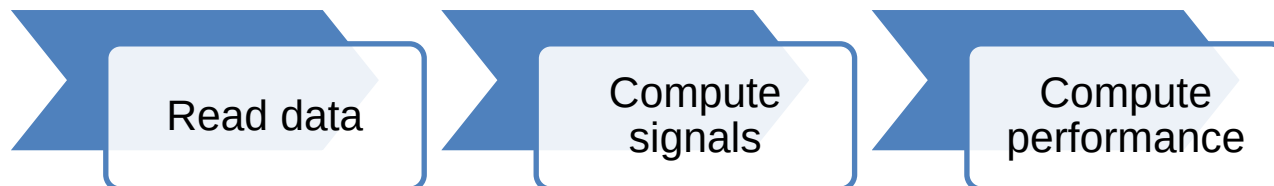
Offer full range of standard API, including supported pipeline API in both backtesting and live trading.

5. Software Engineering Taken Care Of

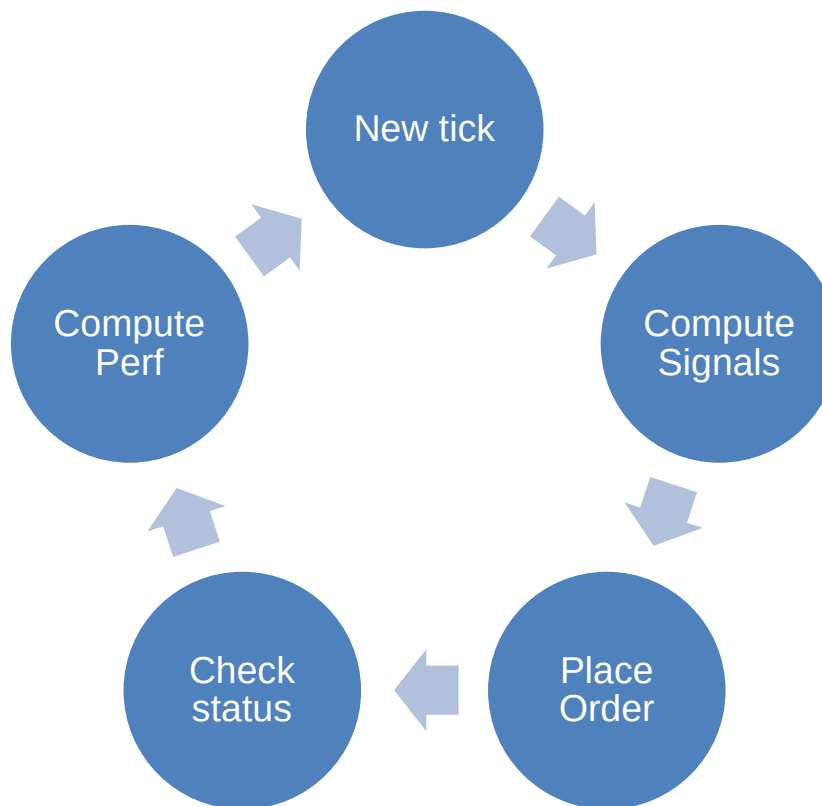
More than **95%** of event driven code is NOT related to the strategy. Blueshift takes care of connection drop, unexpected error, CPU monitoring, risk management etc. Follows a mix of **fail-first** (at start) and **fail-safe** (after run-ins) methods.

Why Blueshift

Vectorized



Event-driven



Why Blueshift



Vectorized Backtesting in fast, but inaccurate

1. Computing prices/ signals ahead of time is a recipe for look-ahead bias
2. Very difficult (often impossible) to include complex state dependency
3. Assumes capital is linearly scalable - 10K vs 10M has very different outcome in reality

Event Driven – Accuracy for Speed

1. Accuracy in execution simulation (slippage models, margin models)
2. No look-ahead guaranteed, by design
3. Backtests are simulated just like live trading – except compressed in time

Highly Optimized, Highly Flexible

Use Apache Arrow (Pandas only when required). Use optimized “sliding” caching. Coded in Cython for CPU-intensive parts

Offers a flexible, extensible API – customizable Slippage model, Margin model and even Broker interface.

Why Blueshift



An Example – CEP vs Vectorized or Rule Engine

EASY: Go long X if SMA1 greater than SMA2

HARDER: Go long X if SMA1 greater than SMA2, if I have a position in stock Y

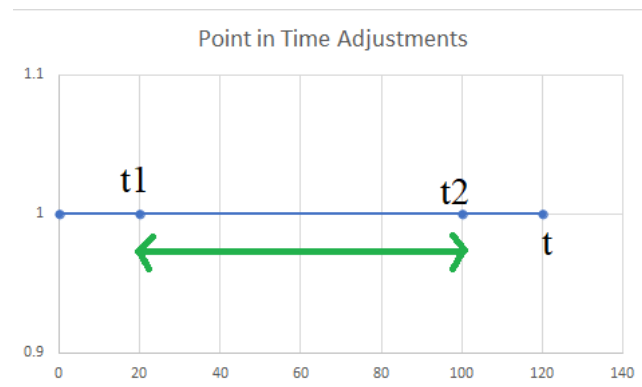
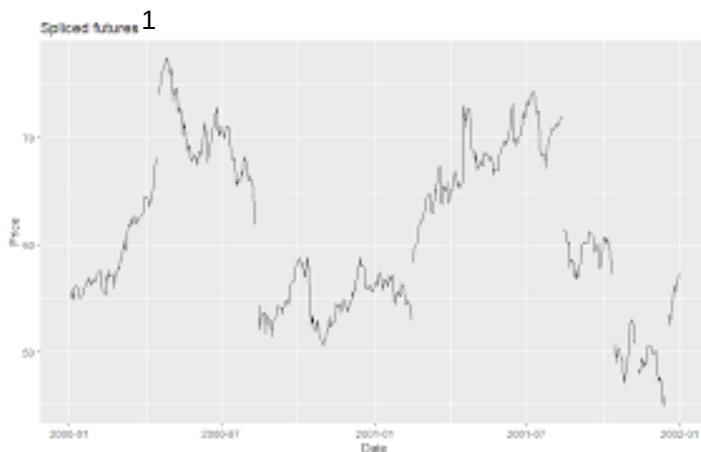
EVEN HARDER IN VECTORIZED OR RULE ENGINE: Go long X if SMA1 greater than SMA2. Track our net pnls. If we detect a change in pnl momentum, exit the strategy

RE-WRITE MOST OF THE CODES FOR EACH NEW STRATEGY

PLUS RE-WRITE AND TEST THE WHOLE CODE FOR EACH DEPLOYMENT

Why Blueshift

An Example – Accurate Historical Data Fetch (Point in Time)



If you query for ACC first futures prices on a particular date-time for last n bars:

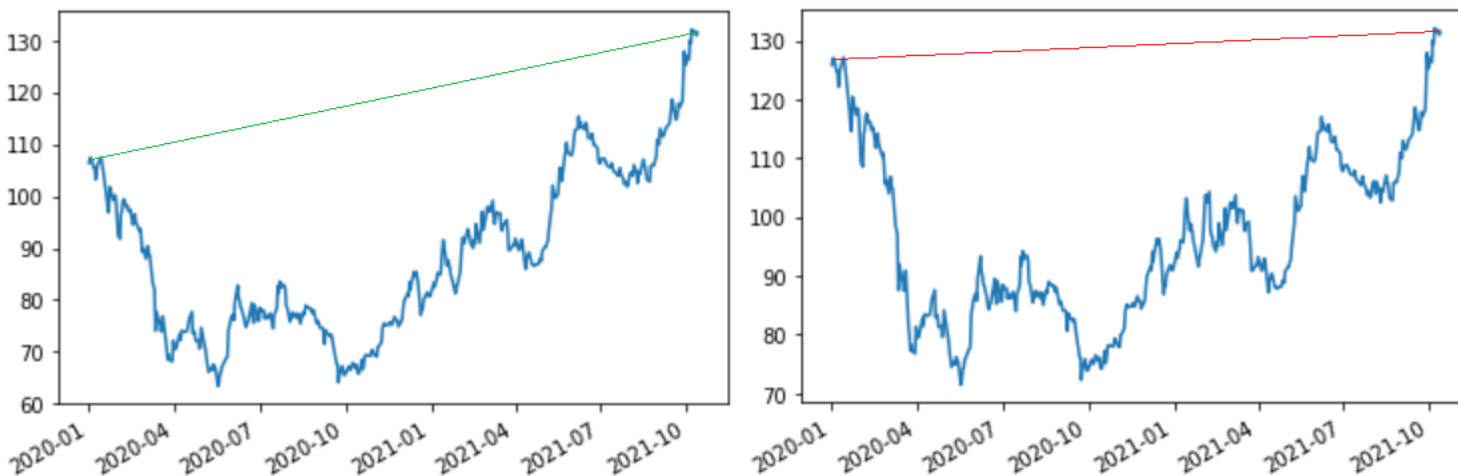
1. Blueshift fetches all the contracts prices in the range
2. Computes backward ratio adjustments for each roll date in the range
3. Apply the adjustments to the raw price series

READ Ops - one each for all instruments in range 2. Dynamically compute cumulative futures ratio between t2 and t2. 3. Dynamically compute corporate adjustments. Apply

1. source: Vojtko, R and Padyšák, M, Continuous Futures Contracts Methodology for Backtesting (January 11, 2020). Available at SSRN.

Why Blueshift

An Example – A Dividend Paying Stock



Applying technical or statistical inference on the price series on the right is quite misleading. The difference? A proper handling of dividends

“Everybody should learn to program a computer,
because it teaches you how to think.”

Steve Jobs (Apple Co-founder)

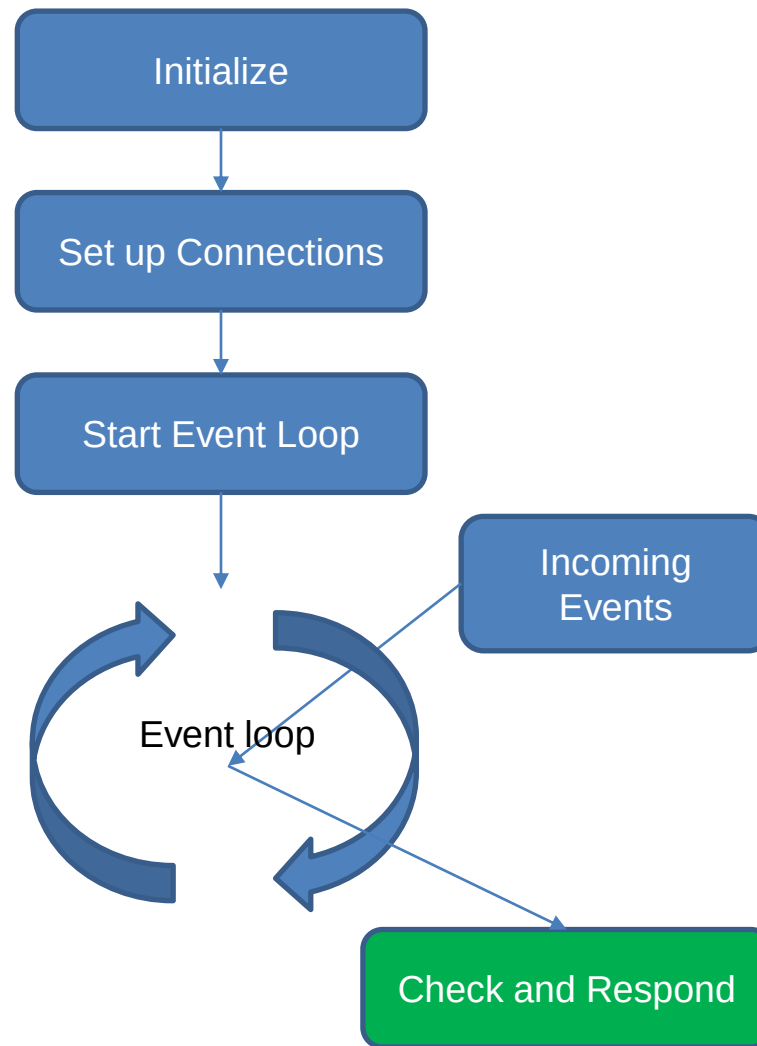
Event Driven Systems

Interacting with Event Driven System

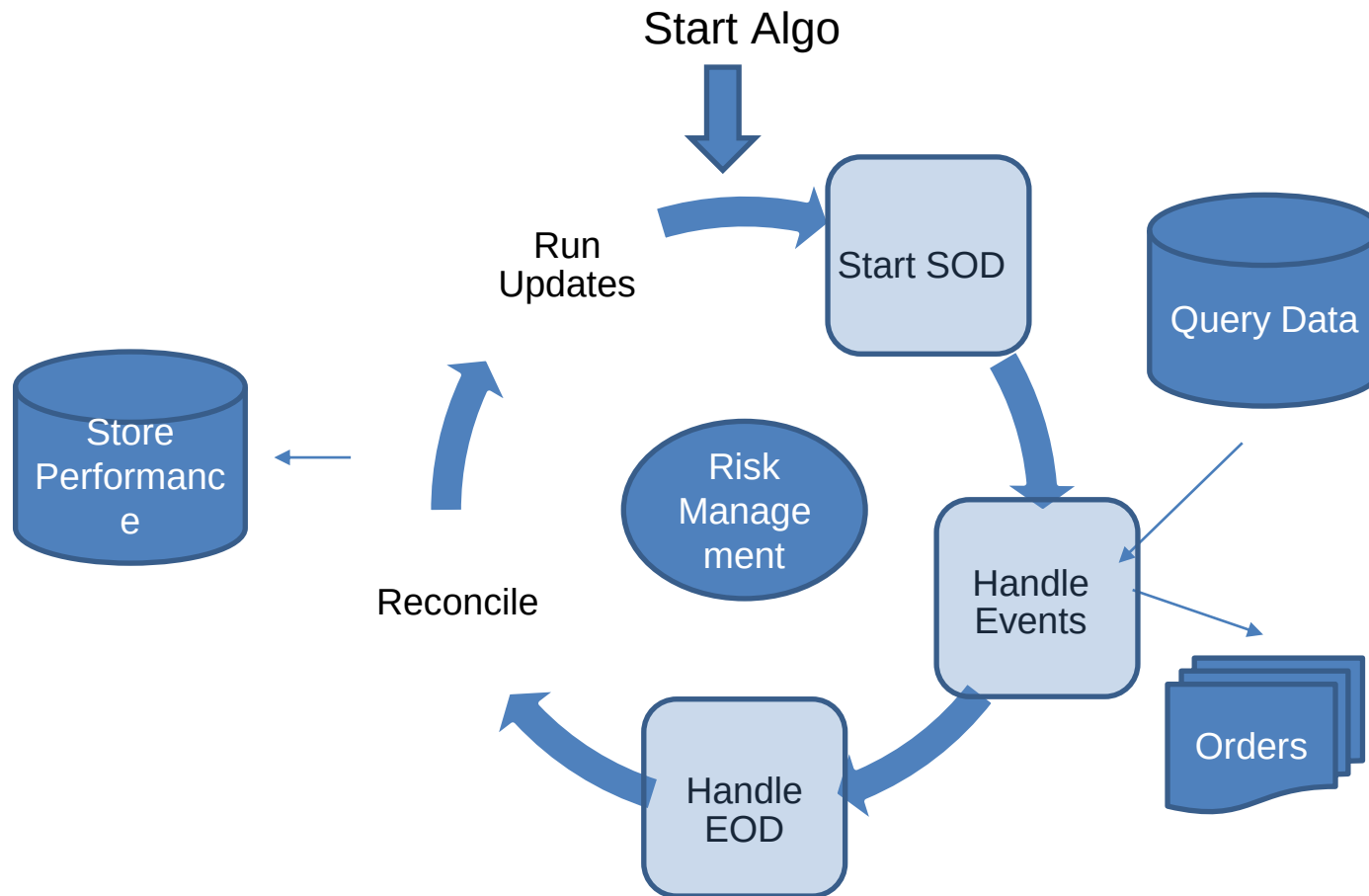


blueshift

Trading System – Big Picture

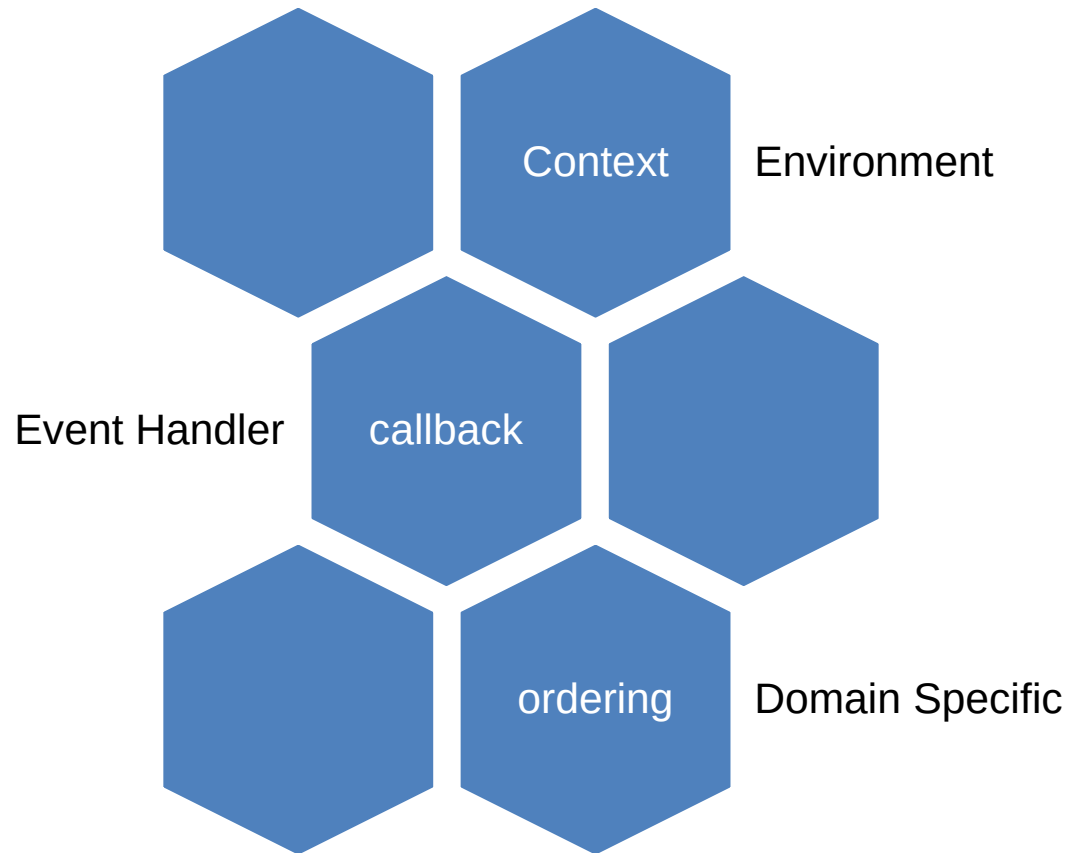


Trading System from Users POV



A trading algorithm is essentially a process that consumes data and generates orders. Other things are additional but very important paraphernalia (risk and capital management, performance tracking etc.)

Interaction With Event Loop



Imperative Vs Reactive Algo

```
from my_trading_systems import *

initialize_database()
initialize_api()
initialize_assets()
initialize_algo_parameters()

while True:
    try:
        current_positions = fetch_current_positions()
        open_orders = fetch_open_orders()
        prices = fetch_data()
        signals = generate_signals(prices)
        positions = generate_target_positions(
            signals, current_positions,
            open_orders)
        orders = calculate_new_orders(positions)
        for order in orders:
            place_order(order)

        open_orders = fetch_open_orders()
        closed_orders = fetch_closed_orders()
        reconcile_orders()
        reconcile_positions()
        pnls = compute_pnls(current_positions)

        if pnls < STOPLOSS or pnls > TARGET:
            exit()

        if end_of_day():
            exit()
    except Exception as e:
        handle_errors(e)

close_database()
close_api()
compute_performance()
save_results()
```

```
from my_event_trading import *

def initialize(context):
    initialize_algo_parameters()

def handle_data(context, data):
    prices = data.fetch_data()
    signals = generate_signals(prices)
    positions = generate_target_positions(
        signals, context.current_positions,
        context.open_orders)
    orders = calculate_new_orders(positions)
    for order in orders:
        place_order(order)

    if context.pnls < STOPLOSS or context.pnls > TARGET:
        exit()
```



Blueshift Event-loop Interaction



Environment

Context: `portfolop`, `orders`, `trading_calendar`, etc.

Setups `set_commission`, `set_slippage`, `set_stoploss`, `ris controls` etc.

Miscellaneous: `get_datetime`, `can_trade`, `record`, `terminate`, etc.

Event Handlers

Time-based events: `initialize`, `before_trading_start`, `handle_data`,

Scheduled Events: `schedule_function`, `schedule_later`

Realtime event handlers: `on_data`, `on_trade`

Domain Specific

Asset fetching: `symbol`

Data fetching `data.current`, `data.history`

Placing orders: `order`, `order_target`, `cancel_order`, `square_off` etc.

“Unfortunately for the case of sanity, that (momentum investing) seems to be true.”

Cliff Asness (AQR Capital)

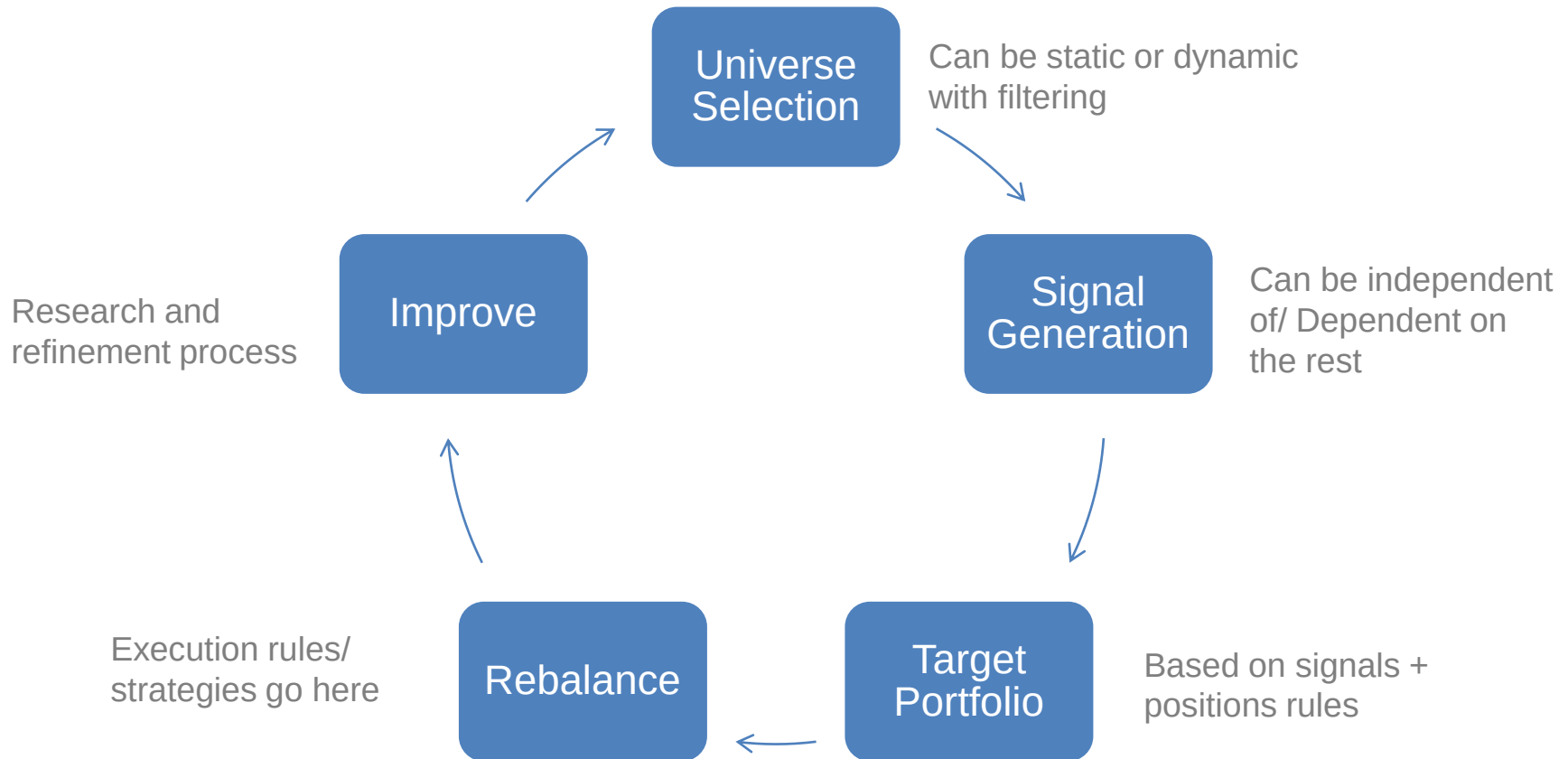
Strategy Work Flow

The General Workflow



blueshift

Systematic Strategy Design Cycle



Systematic Strategy Development



- **Inputs:** Input to the strategies can come in many flavours¹ like
 - Price/ returns and it's transformation. Most of the common technical indicators are transformation of price returns²
 - Positioning information – volumes and open interest data and participant-wise positioning data if available
 - Fundamental information –macro-economic information, company fundamentals
 - Non-market information: Example twitter sentiments, analysts ratings etc.
- **Trading Rules/ Logic:** Can be either based on trader's hypothesis or inferred (learned) from data
 - Form hypothesis (e.g. moving average cross-over signals change in trends) and test – traditional approach
 - Feed data to infer rules (the subject matter of machine learning and artificial intelligence)
- **Back-testing and Forward-testing:** A crucial step and often over-looked step.
 - A good platform to guard against biases : data-mining bias, survivorship bias, market-impact modeling, look-ahead bias
 - Should be flexible, event-driven (to avoid look-ahead bias) and with built-in analytics
- **Portfolio Creation/ Optimization:** Never go all-in with a single strategy
 - Two strategies better than one – if they are uncorrelated (in terms of signals and/ or performance)
 - Various methods exists – ensemble, traditional portfolio theory, dynamic portfolio allocation etc.

1. Ideally sources should be disparate, uncorrelated and relevant.
2. For further technical details see [here](#).

“Unfortunately for the case of sanity, that (momentum investing) seems to be true.”

Cliff Asness (AQR Capital)

Blueshift Work Flow

Developing Strategies



blueshift

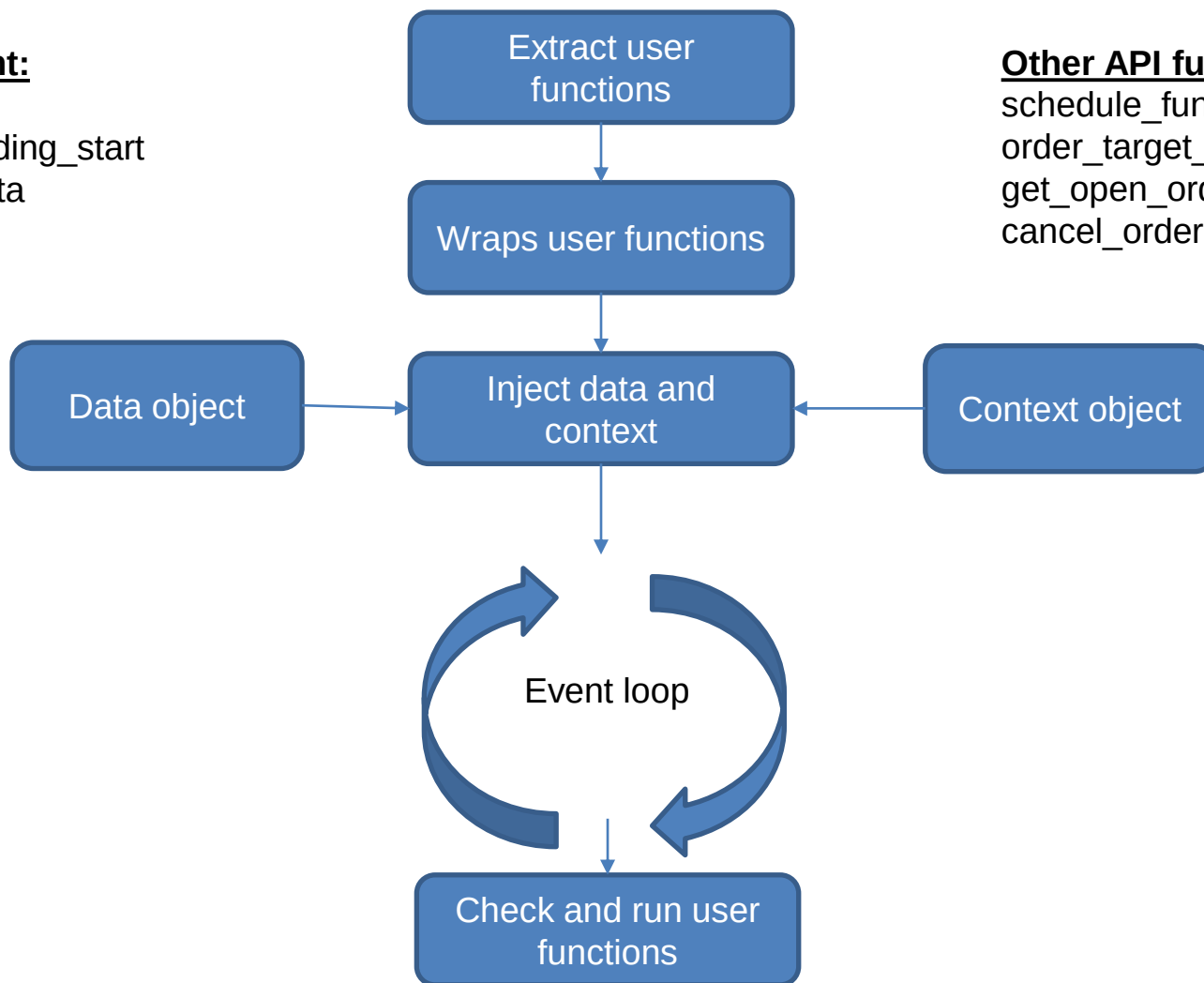
How Blueshift API works

Entry Point:

Initialize
before_trading_start
handle_data
analyze

Other API functions

schedule_function
order_target_percent
get_open_orders
cancel_order



Work-Flow on Blueshift

initialize()

Called once at the start of the algorithm

Place to set up algorithm parameters and initialize them

Arguments: context – a dictionary object to store all user parameters and variables and available to all other functions

before_trading_start()

Called once at the start of the trading session (i.e. once a day)

Place to set up daily pre-open stuff

Arguments: context – as above, and data – a object with current date-time, can be queried for current or historical data on symbols

handle_data()

Called once at every bar (every minute)

Place to set up logic on how to respond to market data

Arguments: context – as above, and data – as above

schedule_function()
date_rules()
time_rules()

order_percent_target()
get_open_orders()
cancel_order()

set_commission()
set_slippage()
set_account_currency()

Other Useful API Functions

Schedule any user-defined functions at a given time and frequency

Functions to place, and cancel orders

Functions to set trading and other parameters

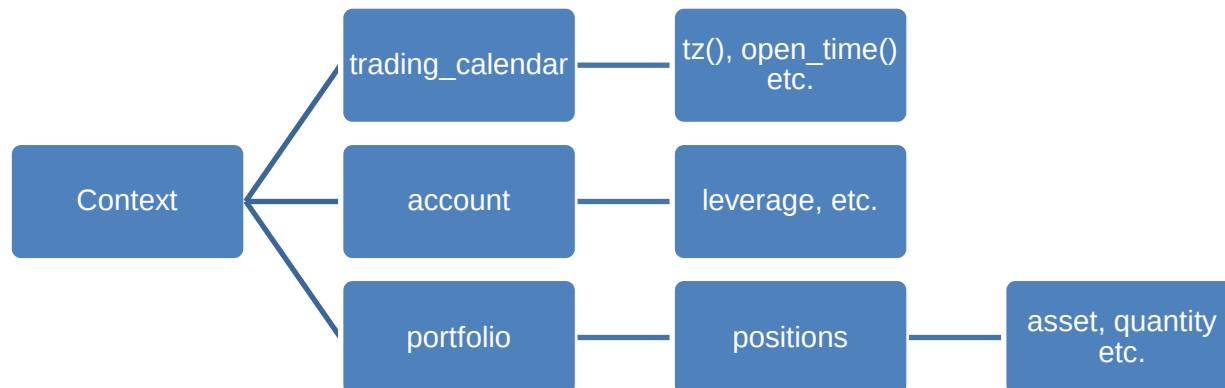
Blueshift Environment



Data contains everything needed to feed data: `current()`, `history()`, as well as `current_dt`

Context contains everything else: Portfolio (`context.portfolio`), account (`context.account`), calendar (`trading_calendar`)

See *here* for more details



Blueshift APIs



Algo callback and scheduling: initialize, handle_data, before_trading_start, schedule_function (with date_rules and time_rules)

Fetch assets: symbol

Fetch data: data.current, data.history

Ordering management: placing orders -> order (order, order_percent, order_value), idempotent order (order_target, order_target_percent, order_target_value), canceling orders – cancel_order, fetch all open orders -> get_open_orders, fetch order by ID -> get_order

Context queries: get_open_positions, context.account, context.orders, context.open_orders, context.portfolio.cash, context.portfolio.positions, context.portfolio.pnl, context.account.gross_leverage, and many more

For full reference – see <https://blueshift.quantinsti.com/api-docs/>

Robust Algos

A computer program does what it is told, it does not know right from wrong

1. **Code Review and Testing:** *Does the code do what we intend to? Or have we left things to assumptions? Look for infinite loops, machine-gunning, corner cases. Test thoroughly, along with code walk-through*
2. **Validate and sanitize:** *Does the combination of input parameters make sense? E.g. Wrong assets, quantity vs lot, lookback vs signal parameters. Does the data look fine? Data can have missing values, or even missing assets/ columns.*
3. **Intention vs the reality:** *An order placement does not mean we have traded. An order cancellation does not mean it is actually cancelled instantaneously. A custom model estimation may not converge.*
4. **Respect Murphy's Law:** *Be aware of unexpected errors – slow internet, out of memory, timeouts etc. Check if in such cases your algo needs explicit error handling.*

Blueshift Workflow Demo



1. *Signing up and Platform Tour*
2. *Creating Strategy*
3. *Running Back-test with a template*
4. *Adding some library functions*
5. *Using our library*

A Simple Strategy



Buy (sell): if close near lower (upper) bands

Buy (sell): if close away from extremes and positive (negative) momentum

“A rising market and a long position is a sure sign of investment genius”

John Kenneth Galbraith (Economist)

Alpha Strategies

Understanding Source of Risks and Returns



blueshift

**QUANT
INSTI**
GO ALGO

Panel 1:

I'VE GOT A STOCK HERE THAT COULD REALLY EXCEL

REALLY EXCEL?

EXCEL?

SELL?

SELL?

SELL?

SELL! SELL! SELL!

KAL '89

Panel 2:

SELL! SELL! SELL!

THIS IS MADNESS! I CAN'T TAKE ANYMORE

GOOD BYE!

GOOD BYE!

BYE?

Buy?

Buy!

Buy!

Buy!

Buy!

Buy!

I'VE GOT A STOCK HERE...

Asset Pricing: Basics

Fundamental Value of An Asset :The value of the asset such that it gives owner a **reasonable profit**¹

Market Price: The price at which (a large number of) players **can agree** to buy or sell an asset.

Question: What is **the fundamental Value** of any thing? Expected Pay-off?



A Simple Game of Luck²:

You pay \$X dollars to play a game against the “house”. The game starts with \$1 in the pot. At each turn the house toss a fair coin. If it is tails, you get whatever in the pot, else we move to the next turn with the house doubling the pot amount (i.e. \$4, then \$8 then \$16 and so on).

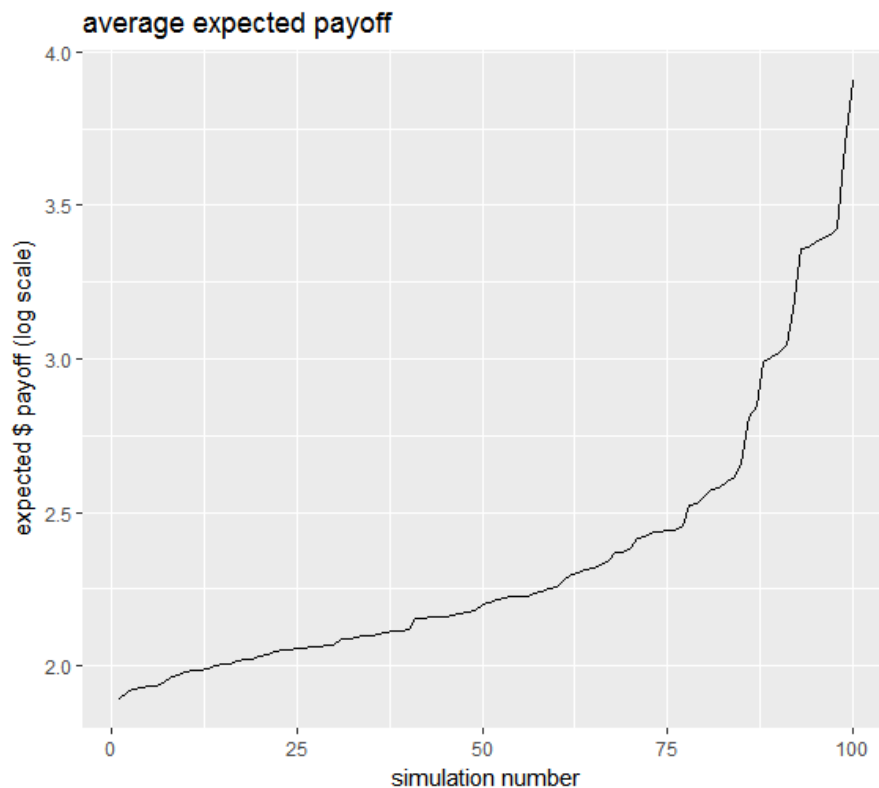
What is X. How much you will pay to play this game?

1. In line with the definition from Adam Smith, Wealth of Nations (1776)
2. Also known as St. Petersburg paradox

Asset Pricing: Paradox

Question: What is **the fundamental Value** of any thing? Expected Pay-off?

Unbounded Expectation!!



1. 100 runs of simulations with 50K tosses each

Asset Pricing: Concepts



Utility: the pleasure or satisfaction derived. A mathematical concept (a function) to arrange a set of choices in a given time and/ or over time in order of preference.

Value of anything is the present expected utility (Expected Utility Hypothesis)¹

Utility brings in to the picture the risks preference as well as time preference

Present Value Models: Fundamental value is **Net Present Value** (NPV) or **Discounted Cash Flow** (DCF) appropriately adjusted for costs (including opportunity costs) and risks.

1. Time Value of Money: People prefer now than later
2. Risk Aversion: People prefer less risky option when the pay off is same
3. Assumes functional capital market (lending and borrowing)

Costs: Usually zero for financial assets

Risk Free Rate: Captures the opportunity costs (in a sense captures temporal preference)

Risk premium: Captures risk aversion – how much investors demand to be compensated for risks in equilibrium

Discount Factors: Include both risk premia and time premia. Depends on individual utility, but in presence of capital market converges to give **“Law of One Price”**

1. The alternative (arguably more realistic) hypothesis is the Prospect Theory
2. Also known as St. Petersburg paradox

Asset Pricing: Maths

Asset Prices: Payoff under different scenarios (s) multiplied with some “**discount factor**”, that takes in to account utility

$$P = \sum_{s=1}^S x^s \varphi^s$$

If $x = 1$ for any scenario s , then the asset is **risk-free asset**. Obviously

$$P^0 = \varphi^0 = \sum_{s=1}^S \varphi^s, \text{ and } R_f = \frac{1}{\varphi^0}$$

Risk free rate

Alternatively we can write

$$P = \sum_{s=1}^S x^s \varphi^s = \varphi^0 \sum_{s=1}^S x^s \frac{\varphi^s}{\varphi^0} = \frac{1}{R_f} \sum_{s=1}^S x^s q^s$$

Risk neutral pricing

Again, alternatively

$$P = \sum_{s=1}^S x^s \frac{q^s}{R_f} = \sum_{s=1}^S p^s x^s \left(\frac{q^s}{R_f p^s} \right) = \sum_{s=1}^S p^s x^s m^s = \mathbb{E}(mx)$$

Expected “utility” angle

Risk Premium: From basic stats we get

$$P = \mathbb{E}(mx) = \text{cov}(m, x) + \mathbb{E}(m)\mathbb{E}(x)$$

Re-arrange this as

$$\mathbb{E}R = R_f - R_f \text{cov}(m, R), R = \frac{x}{P}$$

Risk premium

Asset Pricing: Theories

Arbitrage Pricing Theory (APT):

$$\mathbb{E}R = R_f - \sum_{i=1}^K \beta_i \left(R_f \mathbb{E}(m \cdot F_i) \right), \text{ where } m = \frac{1}{R_f} + \sum_{i=1}^K b_i F_i$$

Here **F** are the factors and beta are the factor exposure of that particular asset

Capital Asset Pricing Model (CAPM)

This assumes there is only one factor, the market exposure: High returns because higher risks!

$$\mathbb{E}R = R_f - \beta \cdot \mathbb{E}R_M$$

Fama French 3-Factor Model

Some other factors than market, like market capitalization (size factor) and book-to-market ratio (value factor) are important too!

Note:

- These models are useful for portfolios (large N, like investor B). We are ignoring idiosyncratic risks based on the assumptions that they can be diversified away!!
- These are relative valuation (relative to factors), NOT absolute valuation (NOT how much is Infosys worth to you, but what the value of average investors put on it)
- Not useful for pricing derivatives, hard to find stable factors (or appropriate discount rates)! Rather price in terms of the underlying instead (risk neutral pricing)

What is Alpha

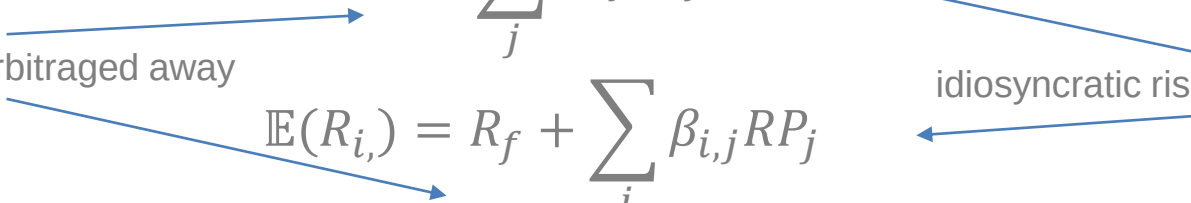
Modern Asset Pricing Theories are essentially as follows:

$$R_{i,t} = a_i + \sum_j \beta_{i,j} \Delta F_{j,t} + \epsilon_{i,t}$$

Alphas will be arbitrated away

$$\mathbb{E}(R_{i,t}) = R_f + \sum_j \beta_{i,j} RP_j$$

idiosyncratic risks not paid for



1. ΔF is k -variate $I(0)$, ϵ is n -variate white noise (total instruments n , total factors k). RP s are risk premia (k -variate $I(0)$)
2. Objective: Create portfolio to beat the market!
3. First try: Capture any factor other than market – Jensen's alpha!
4. Holy Grail: RP (and ΔF too!) not time-dependent – pure alpha!
5. Tough luck: try RP not time-dependent – Beta is the new alpha?
6. Next: RP is stable and consistent – Focus on factor allocation!
7. Time-varying but predictable? – Focus on market timing!
8. Unstable and unpredictable on all scale? – go back to picking stocks!

“Successful investing is anticipating the anticipations of others.”

John Maynard Keynes (Economist)

Systematic Equities

A very brief overview



blueshift

Momentum: Concept



What is Momentum: Exploiting time series persistence

Strategy-I: Buy (sell) assets that have been going up (down) in prices in the past, hold them in the portfolio for a defined period of time (holding period), and then rebalance. This is known as time-series momentum

Strategy-II: Rank all assets based on their past returns. Buy the top x-percentile and sell the bottom x-percentile (even if they are going up in prices). Rebalance after the holding period

Systematic Asset Management Before Momentum:

1. Passive - Portfolio Optimization: $\min(w^T \cdot \Sigma \cdot w - q \cdot R^T \cdot w), s.t. q \geq 0, \sum_i w_i = 1$
2. Active - Market Timing and signal generation (digital signal processing, technical analysis etc.)

Systematic Asset Management after: Factor investing:

1. Passive - Portfolio Optimization (mean-variance)
2. Passive – Portfolio Optimization (risk factors optimization)
3. Active - Market Timing and signal generation (digital signal processing, technical analysis etc.)

Momentum: Everywhere



Equities: classic cross-sectional and time-series momentum (we have just seen it)



Fixed income: Central banks tend to hike (cut) rates in steps, leading to momentum effect on short term rates. Also economies undergoes business cycles that get reflected in rates and inflations – leading to trends in long term rates



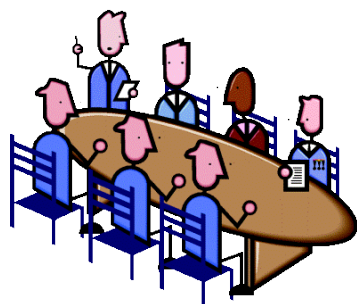
Commodities: One of the first markets to trade momentum (pioneer of technical analysis). A strong drivers are business cycles and persistence of under or over capacity (setting up factories require time). Tend to mean revert over long period.



Foreign Exchange: A large part of momentum is driven by the same factors as rates.

Momentum: Drivers

Behavioural



Under-Reaction: Slow reactions to company specific or macro-economic news (the opposing theory is known as Efficient Market Hypothesis or EMH). Conservative behaviour, bounded rationality, liquidity constraints etc.

Delayed Over-Reaction: Behavioural feedback mechanism. This also gets reinforced by under-reaction

Disposition Effect: Investors tend to stick to losers too long and book profit too quick. Creates momentum.

Risk-based



Risk Compensation: Market value of firm growing faster than fundamental increases the risk, hence momentum profit should compensate. Momentum is usually stronger for companies with higher growth opportunities and riskier cash flows

Correlation Structure: Momentum assets strongly correlate with each other, which implies a common factor which can experience changes in future (**momentum crash**)

Liquidity Risks: Recent winners have greater exposure to liquidity risks (everyone tries to exit at the same time)

Blueshift Pipelines



Pipelines are efficient ways to compute a factor on a large universe

*We attach a pipeline at the onset (initialize). This is `attach_pipeline()`
Then call the results periodically: `pipeline_output()`. This returns usually a pandas dataframe with multi-index (indexed by stocks and then timestamps).*

To define pipeline output, we define the factor. This is basically defining a class with a ``compute`` function that must return the factor we are interested in. It takes a ``input`` (usually the price), a ``window length`` to compute, and all assets in the database. It returns the computed value

Blueshift Pipelines

Define factors



Compute Dependencies



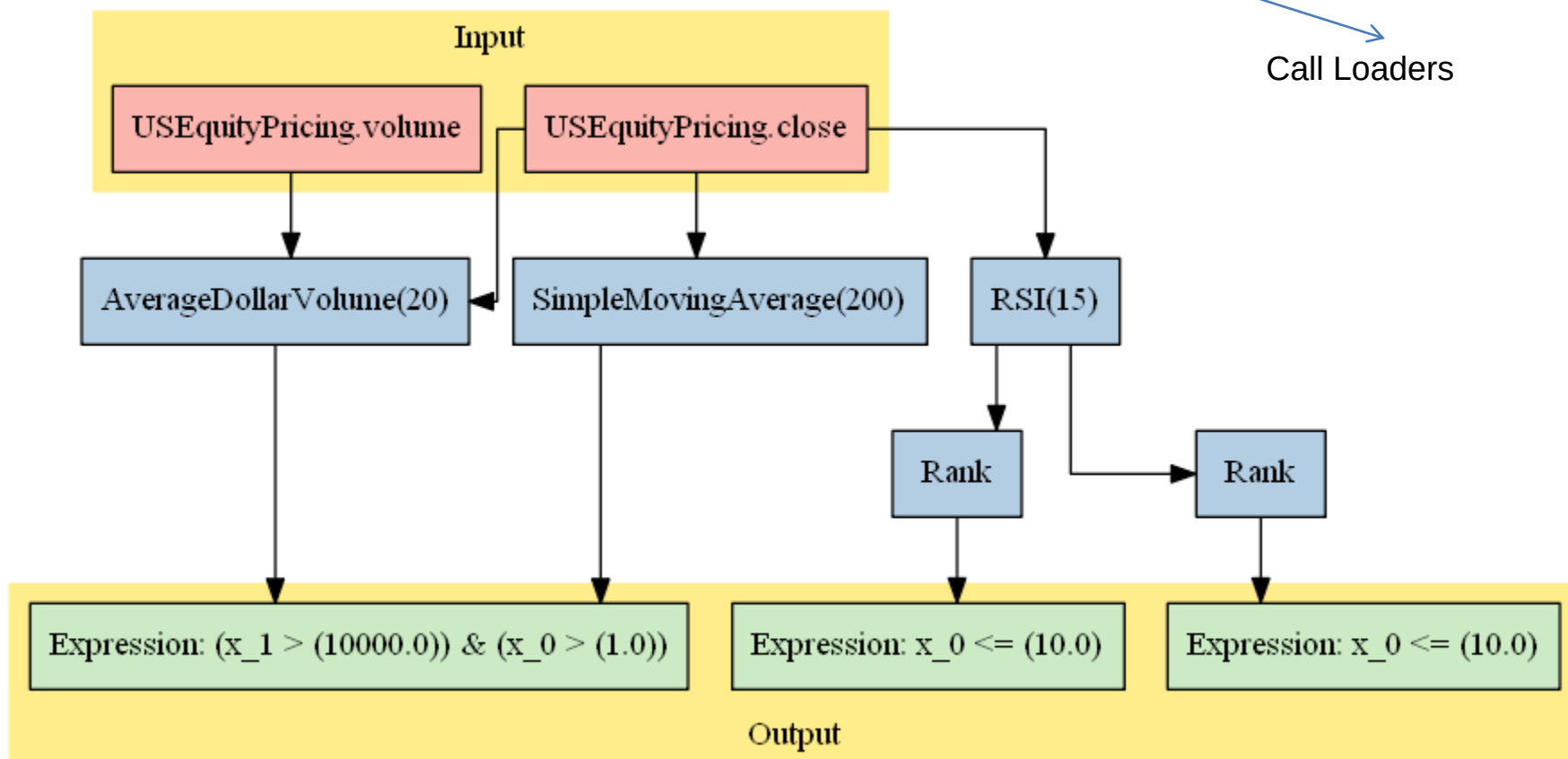
Create Graph



Topological Sort



Call Loaders



Screen

Columns

Blueshift Workflow Demo

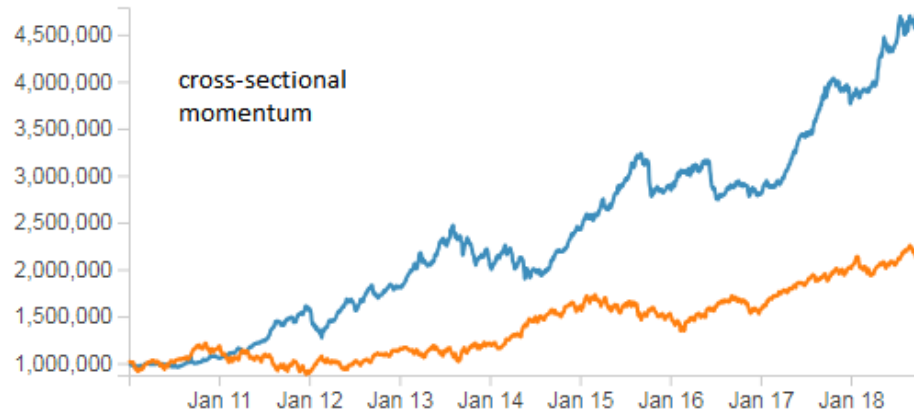


1. *Create a library for momentum factors*
2. *Import that to develop a cross-sectional strategy*
3. *Use same library to develop a time-series strategy*
4. *Compare results*

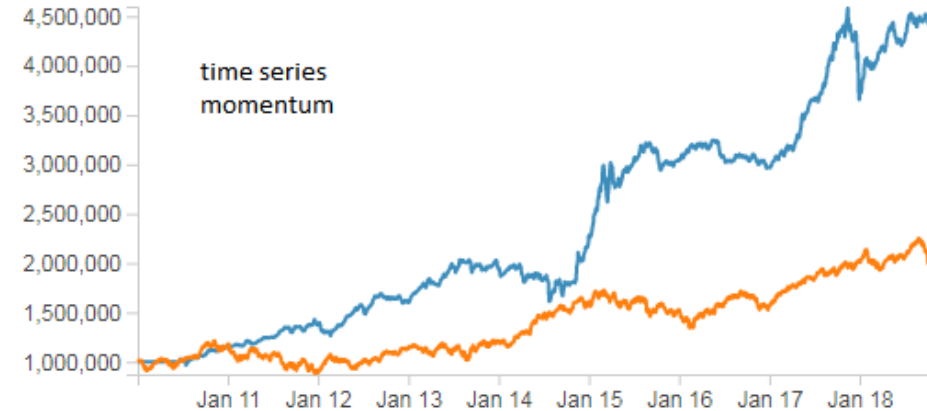
Momentum: Performance

RETURNS	ALPHA	BETA	SHARPE	DRAWDOWN	RETURNS	ALPHA	BETA	SHARPE	DRAWDOWN
366.54%	0.19	-0.14	1.55	-24.06%	344.00%	0.17	-0.04	1.53	-21.89%

■ PortfolioValue ■ Benchmark



■ PortfolioValue ■ Benchmark



1. The side-ways markets during 2011-2013 saw better performance from XMOM
2. Strong post-election rally was better captured in TSMOM
3. XMOM had a shorter “flat” period post election rally
4. In general, “beta” is higher for TSMOM (especially for absolute beta)
5. Overall performance, including alpha, Sharpe ratio, drawdowns are similar

Let's see the demos

1. Based on NSE 500 stock universe, with look back period of 12 months and rebalance period of 1 month

“Trial and error is freedom.”

Nassim Nicholas Taleb

Statistical Arbitrage

Statistics, and Practice



blueshift

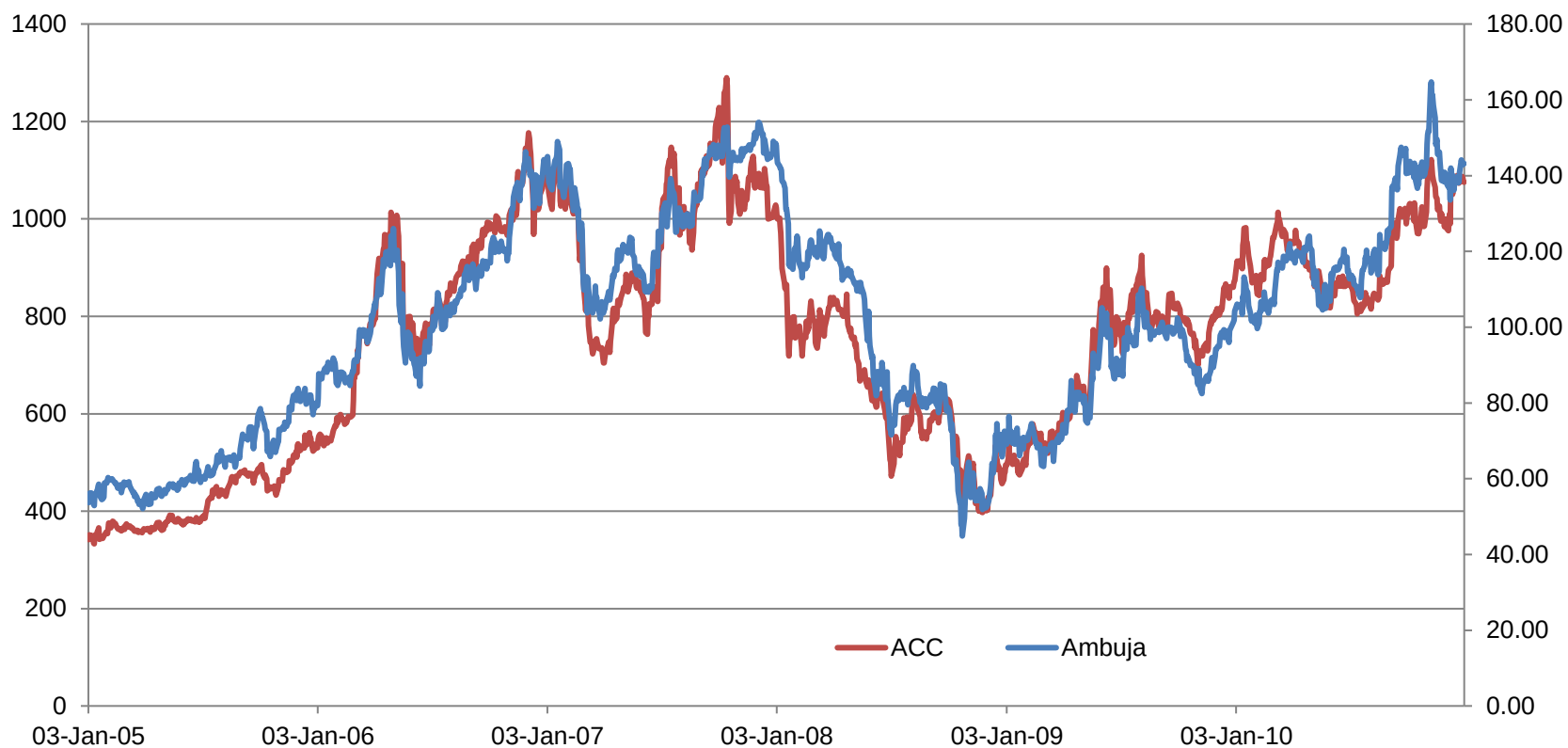
Stat Arb: Concepts

- Choose a pair of stocks that move together very closely, based on a certain criteria (i.e. TCS & Infosys)
- Wait until the prices diverge beyond a certain threshold, then short the “winner” and buy the “loser” (contrarian approach)
- Reverse your positions when the two prices converge --> Profit from the reversal in trend
- Pairs trading strategy requires few key decision points:
 - **Pair (or basket) Selection Algorithm**
 - **Market Timing (Entry and Exist)**
 - **Optimal Position Sizing Mechanism**
 - **Spotting the Scarce Opportunity**

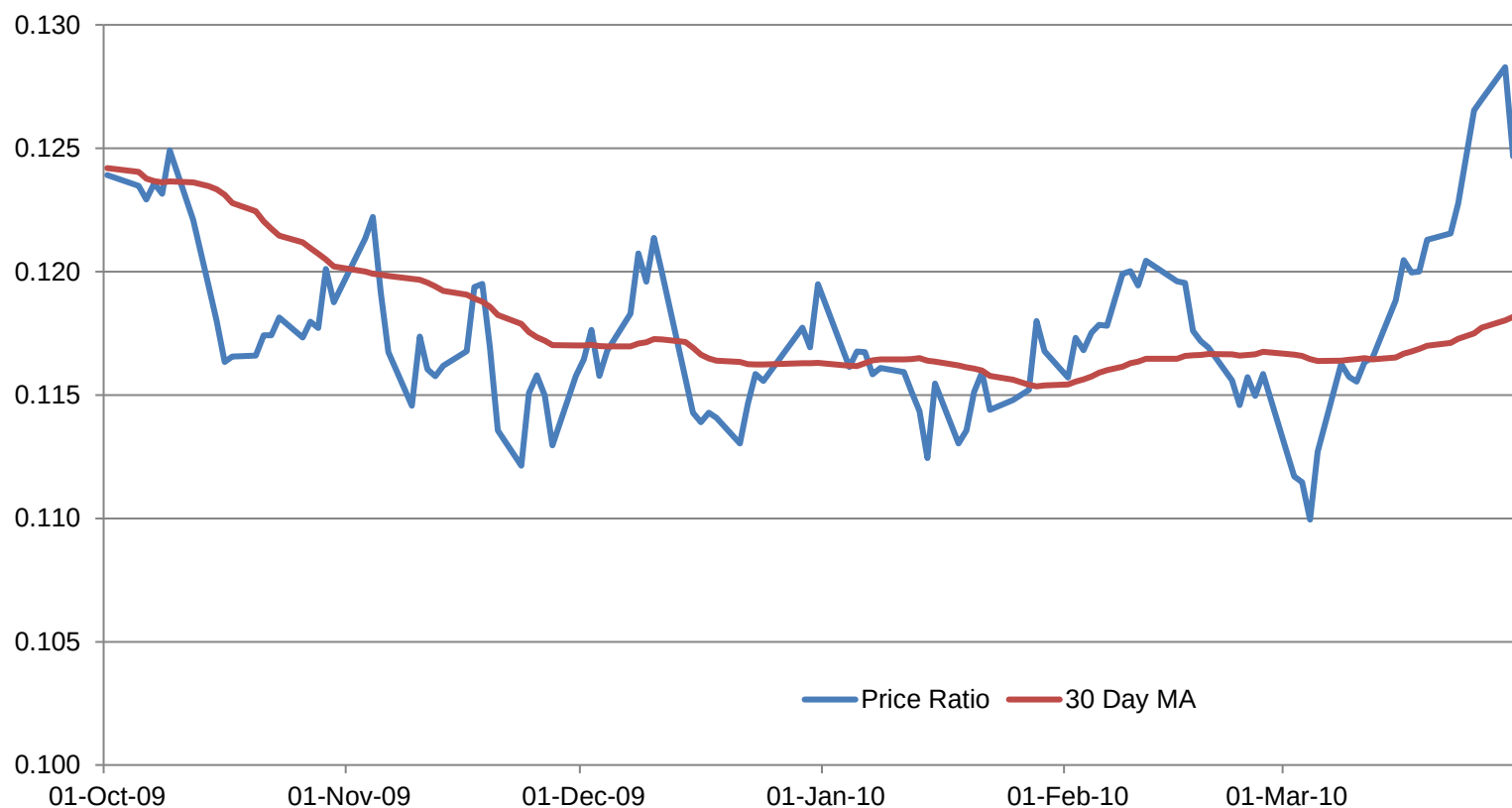
Stat Arb: Risks

- Statistical Arbitrage is not a risk-free strategy. Key risk involved in trading are as follows:
 - Rather than converging, prices of the two securities can begin to diverge (drift apart), i.e. the spread picks up trend rather than mean-reverting to the original mean and the cointegration can be broken.
 - Sometime, an event in a security can trigger extreme movement in pair ratio. Entering into such a security trade should be avoided.
 - Strict risk management of Stop Loss is required to handle adverse situations once the mean-reverting behavior is invalidated.
 - Inherently stat-arbs are like short gamma trades. They tend to win most of the time and sometimes blow up!

Stat Arb: Example



Stat Arb: Example



Stat Arb: Risks

- Statistical Arbitrage is not a risk-free strategy. Key risk involved in trading are as follows:
 - Rather than converging, prices of the two securities can begin to diverge (drift apart), i.e. the spread picks up trend rather than mean-reverting to the original mean and the cointegration can be broken.
 - Sometime, an event in a security can trigger extreme movement in pair ratio. Entering into such a security trade should be avoided.
 - Strict risk management of Stop Loss is required to handle adverse situations once the mean-reverting behavior is invalidated.
 - Inherently stat-arbs are like short gamma trades. They tend to win most of the time and sometimes blow up!

Stat Arb: Cointegration

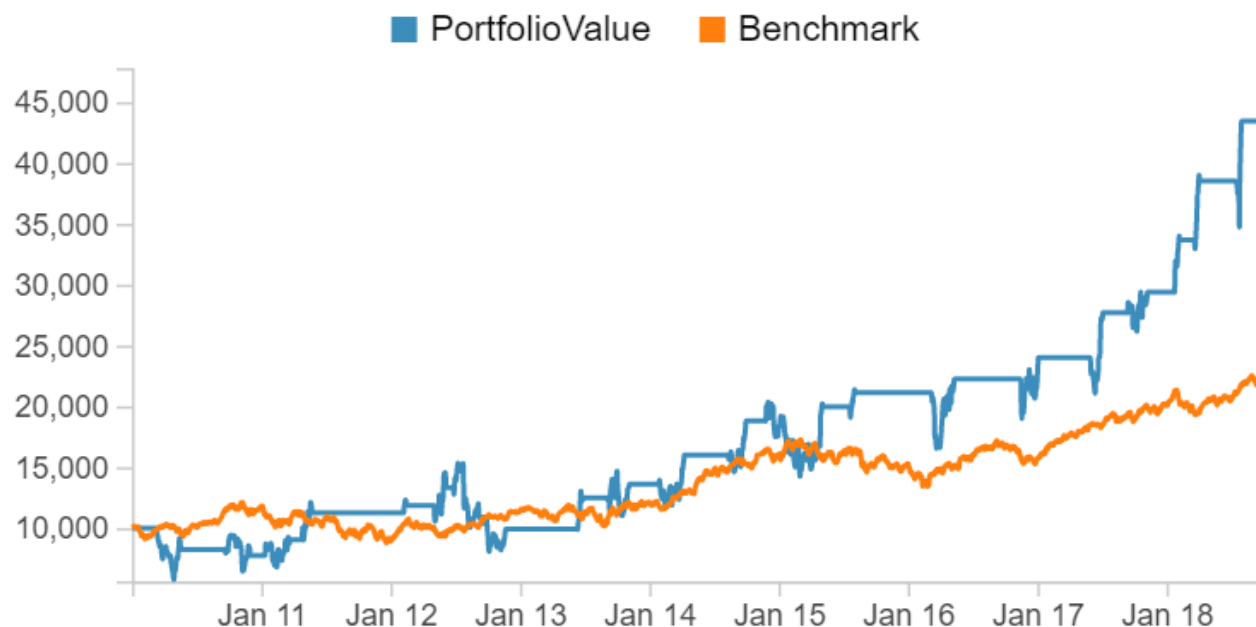
- What is co-integration?
 - If there exists a stationary linear combination of non-stationary asset prices, the combined prices are said to be co-integrated.
- What are the co-integrated process?
 - Cash and Future
 - Stocks in same industry and similar business model
 - Term structure of yields, etc.
- How are they useful for Pair Identification?
 - Return-based pair-trading only works at high frequency!!
- Process of Determining Co-integration
 - Techniques used to identify Co-integration are:
 - Augmented Dickey-Fuller Test
 - Jonhansen Test

Stat Arb: Steps

- Pair Identification
 - Based on prior knowledge
 - Based on quantitative analysis (example: cluster analysis)
 - Finally confirm hypothesis by cointegration test
- Rich cheap identification?
 - Track the cointegrated pair and compute z-spread
 - Define threshold of entry and exit
 - Check for idiosyncratic nature (news flows, company specific announcements etc)
- Risk Management
 - Position sizing: based on expected time to convergence, spread volatility over the period and strength of the signal
 - Step in vs all-in sizing

Stat Arb: Performance

RETURNS	ALPHA	BETA	SHARPE	DRAWDOWN
377.95%	0.26	-0.08	0.63	-49.45%



“A rising market and a long position is a sure sign of investment genius”

John Kenneth Galbraith (Economist)

Systematic Trading in FX

FX Styles and Alpha Strategies



blueshift

The Fundamental Beta in FX



Value: “cheap” currencies will go up in value, and reverse for “expensive” ones. Measure of richness/ cheapness requires economic or empirical models (like purchase power parity, REER or statistical regression-based etc.)



Momentum: Past returns are persistent. Rank currencies on past returns, long the top ones, short the bottom ones (cross-sectional momentum). Alternatively, long if past returns are positive (above threshold) and short if the reverse (time-series momentum)



Carry: Positioning to profit if nothing changes in the universe except passage of time. Long currencies with higher yields (interest rates) and funded by ones with low yields.



Defensive: Low risk currencies (better sovereign ratings) produces better risk-adjusted returns. Long currencies with higher sovereign ratings and short the bottom ones.

Creating A Carry Strategy

1. *A bit more on the logic*
2. *Which one we should buy and which ones to sell*
3. *Coding the strategy – Starting from the template*
4. *Adding the momentum logic*
5. *Adjusting the re-balance function*
6. *Running the back-test*

Creating Other Factor Strategies



1. *Pick up from where we left*
2. *Which one we should buy and which ones to sell*
3. *Coding the strategy – Starting from the template*
4. *Running the back-test*

Creating A Factor Basket



1. *Pick up from where we left*
2. *Simple allocation of $1/3$ in to each of carry, value and momentum*
3. *Combining the strategies*
4. *Running the back-test*

“Past performance is the best predictor of success”

Jim Simons (Renaissance Technologies)

Back-test Evaluation

Maths , Pseudo-maths and Overfitting



blueshift

Risk Reward Metrics

Equity Curve – picture tells a thousand metrics

Algorithm Metrics :

Risks and Performance: Total returns, annualized returns, volatility, Sharpe ratio, Sortino ratio, Omega ratio, time-series stability, skew, kurtosis, max drawdowns, alpha, beta, net and gross leverage,

Trade Metrics: *Average number of trades, per trade profit, Win rate, Total trades, Profit-loss ratio, average wins, average loss, average holding period, average costs, maximum adverse excursion, maximum favorable excursion*

Scenario Analysis: *Strategy performance in defined periods – 2007 quant crisis, 2008 financial crisis, 2011 US Budget crisis, 2010-2012 European sovereign crisis, EM crisis, Other identifiable periods of idiosyncratic risks*

Risk Reward Metrics

Sharpe Ratio:

$$Sharpe = \frac{R_P - R_f}{\sigma_P}$$

Sortino Ratio:

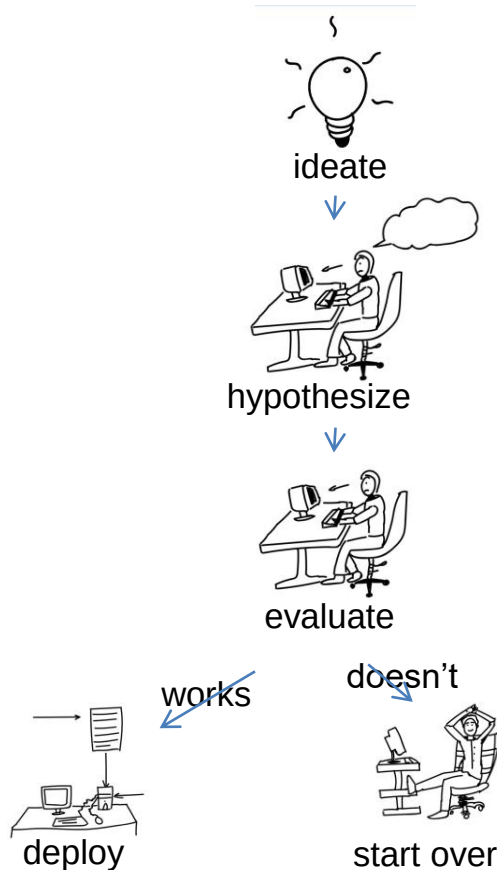
$$Sortino = \frac{R_P - R_B}{\sigma_{down}}$$

Omega Ratio:

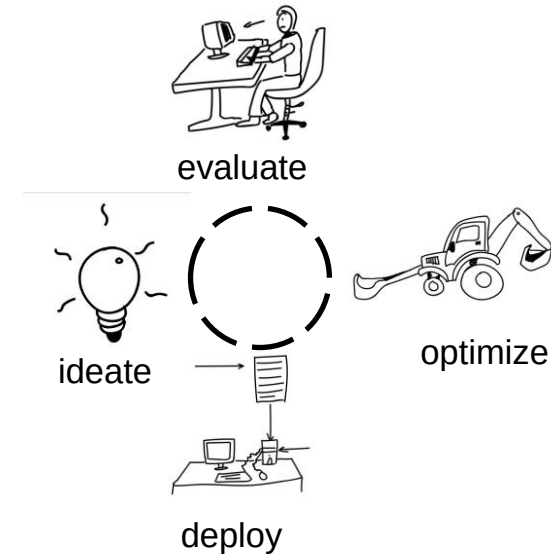
$$\Omega = \frac{\int_r^{\infty} (1 - F(x)) \cdot dx}{\int_{-\infty}^r F(x) \cdot dx}$$

Back-testing: Optimize?

Scientific way of developing a strategy:



The way most of us end up with:

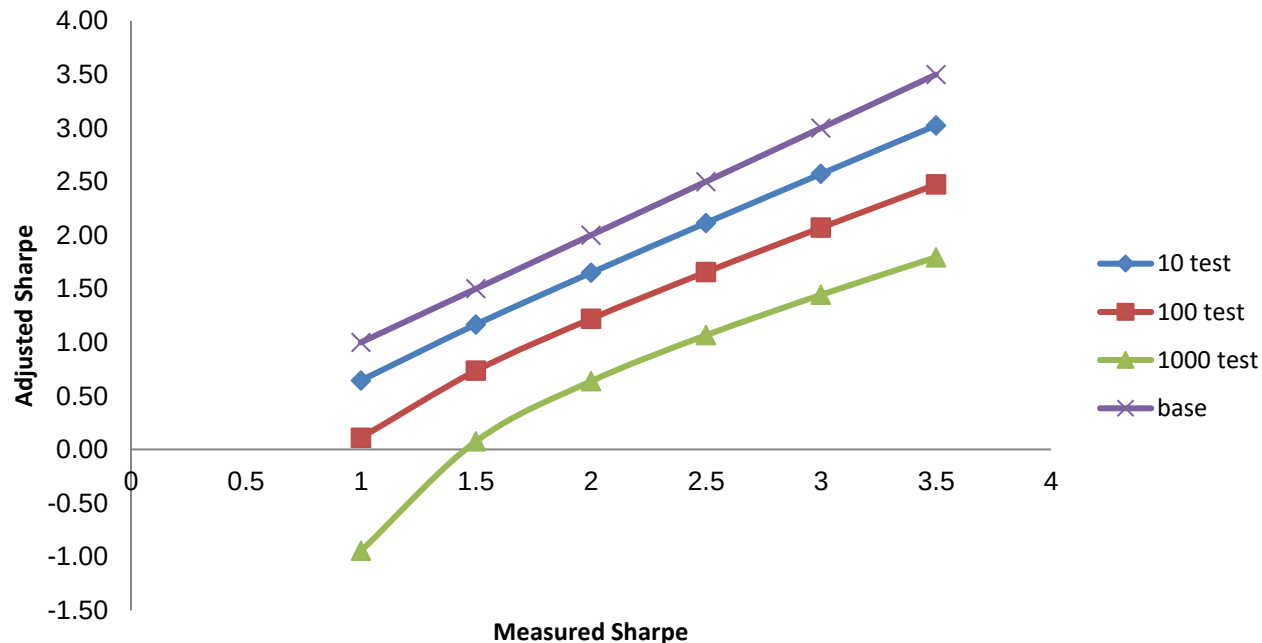


Back-tests are useful for hypothesis testing
Not a data-fitting tool

We do not want to optimize our backtest performance. We want to optimize expected live performance.

Problems with Optimization

Sharpe Ratio Adjustment



Alternative to Optimization

1. Adaptive Strategies: Example Change Point Analysis
2. Stable Factor research: With economic/ behavioural justification
3. Ensemble Strategies: Works best with diverse strategies [not in demo]
4. Validation – cross validation or paper-trading

What It Is Not



Black-box: A systematic strategy should **NOT** be a black-box. We should **ALWAYS** be able to explain why we buy what we buy (and sell); and why does it make money.



Risk-free Profit: No systematic strategy (or any strategy!) is risk free, and will **never** make money all the time, every time. Risk management is very important.



Fire-and-Forget: A systematic strategy is an **evolving** one. It must pass through the constant cycle of performance measure, tuning and re-risking.



Man vs. Machine: Systematic strategies are not about man vs. machine, but man **and** machine. Human brain is far better evolved to develop hypotheses. Machines are far better at testing and executing them.

Strategy Development - Recommendations



*Always save all your strategy variables in the ``context`` environment
- helps tracking variables and re-starts*

*Use `schedule_function` if your algo does not need to respond to every bars
- will run faster*

Good practice to check account leverage to make sure the algo is running as intended

Do not over-fit – build a hypothesis and test. Data-mining (p-hacking) generates good performance in the past and bad going forward

Check the stats in the backtest results. A strategy with low Sharpe, concentrated wins (low stability), large downside risks (negative skew), and large drawdowns and high beta are not one you can actually invest in

List of Demo Strategies

1. *Strategy Template*
2. *Value Strategy*
3. *Carry Strategy*
4. *Time-series momentum strategy*
5. *Cross-sectional momentum strategy*
6. *Factors basket strategy*
7. *Bollinger band based break-out strategy*
8. *RSI based mean-reversion strategy*
9. *Correlation based statistical strategy*
10. *FX daily computation template*
11. *Change point based regime switching strategy*

Find them at: <https://github.com/QuantInsti/blueshift-demo-strategies>



THANK YOU!

For further queries reach out to blueshift-support@quantinsti.com